

Assessing the Influence of Managed Lane Separation Type on Lane Deviation

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Abstract

Managed lanes (MLs) are increasingly implemented to enhance mobility and safety on congested freeways, but researchers have rarely explored how the type of separation between MLs and general-purpose lanes (GPLs) affects lane deviation, a key factor linked to crash risk, particularly sideswipe collisions. This study examines the relationship between ML separation type and lane deviation using naturalistic driving data from the Strategic Highway Research Program 2 in Washington State. The dataset includes 5,976 trips made by 310 drivers traveling along I-5 and I-90, focusing on three separation types: buffer, median, and concrete barrier. An Analysis of Variance was conducted to determine whether there were significant differences in mean lane deviation across separation types. A random forest model was then applied to assess the importance of explanatory variables, followed by a mixed-effects multinomial model to identify factors influencing lane deviation for drivers on ML facilities. Results show that, on average, drivers tend to steer slightly away from the separator across all designs. Median-separated facilities resulted in the highest mean lane deviation compared with concrete barriers and buffer-separated MLs. Separation type, driver age, driver gender, total miles driven, vision acuity, lane width, shoulder width, driver risk-taking behavior, and the number of MLs and GPLs were all significant factors influencing lane positioning behavior at the 95% confidence level. These findings provide valuable guidance for transportation agencies when selecting separation types for ML facilities, helping them improve safety, reduce crash risk, and improve the overall safety performance of MLs.

Keywords

lane deviation, managed lanes, separation type, mixed-effect multinomial model, naturalistic driving study

Background

Managed lanes (MLs) are highway lanes where operational strategies are proactively implemented and managed in response to changing traffic conditions. They are commonly built next to general-purpose lanes (GPLs) and offer various benefits, such as extra travel lanes for longer regional trips, improved transit services, hurricane and emergency evacuations assistance, and better system connectivity between critical limited access facilities (1). MLs have become an integral part of highway systems worldwide, primarily adopted to address congestion and enhance roadway safety.

By separating certain lanes from GPLs, MLs offer transportation agencies a controlled environment to implement tailored traffic management strategies (2–4). The deliberate separation of MLs is fundamental to their effectiveness, allowing agencies to optimize lane usage and achieve their intended mobility and safety objectives (2–4). There are several types of ML separators,

including pylons, concrete barriers, medians, and buffer separations. While concrete barriers and pylons provide some form of physical barrier between the MLs and the GPLs, double solid white lines only provide a psychological barrier between the two types of lanes. Figure 1 presents the different types of ML separators.

Literature Review

The choice of separation type in an ML facility can significantly affect driving behaviors such as lane positioning, lane-changing frequency, speed choice, and lane deviation (5, 6). As a result, understanding these effects is critical to optimizing the design of ML systems. Lane

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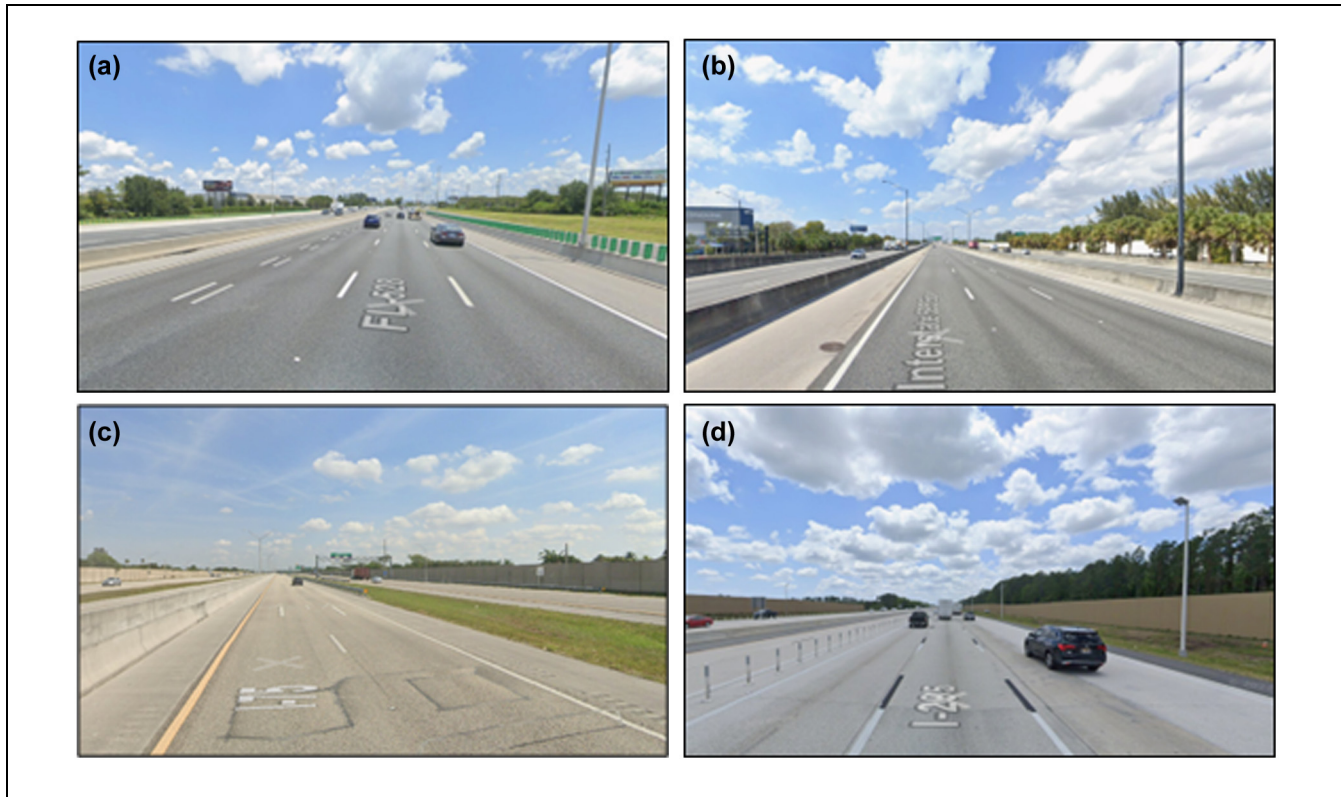


Figure 1. Managed lane separation types: (a) buffer, (b) concrete barrier, (c) median, and (d) pylons.

Source: Google Maps.

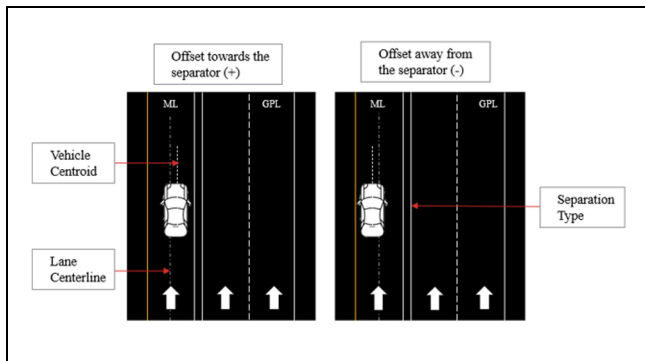


Figure 2. Lane deviation offset.

deviation, defined as the offset between the position of the vehicle's centroid and the centerline of the lane, is a key safety performance measure, as drivers encroaching on adjacent lanes can compromise safety. Figure 2 presents the graphical representation of the lane deviation offset.

Lane deviation, defined as the offset between a vehicle's centroid and the centerline of the lane, is a key metric for assessing safety. Lane deviations are particularly relevant in MLs, as any encroachment into

adjacent lanes can compromise the safety of all road users. The proximity of separators often influences drivers' lane-keeping behavior, with deviations potentially linked to factors such as perceived lane width, separation type, and driver confidence (5). Previous studies indicate that negative lane deviations, where drivers steer away from separators, are common, highlighting the psychological and behavioral impacts of separator design on drivers (7).

Lane deviation in MLs is influenced by a combination of roadway design elements, traffic dynamics, and driver characteristics. Separator type plays a critical role, as different separator choices, such as pylons, buffers, or concrete barriers, can prompt distinct behavioral responses from drivers, influencing how consistently they maintain their position within the lane (8). Likewise, roadway geometric characteristics, including lane width, shoulder width, and the number of lanes, affect how drivers perceive and navigate MLs (9). Narrower lanes or shoulders, for example, may increase driver discomfort, leading to greater deviations (10).

In addition to design considerations, traffic characteristics, such as average annual daily traffic, can reflect congestion levels that may affect driver behavior and lane-keeping precision. Finally, driver demographics and

vehicle characteristics add another layer of variability (11). Factors such as age, gender, ethnicity, and even vehicle class influence how drivers interact with MLs and respond to roadway conditions (8). Together, these variables underscore the complexity of understanding lane deviation and the need for comprehensive analysis to develop safer, more effective ML designs.

Investigating the impacts of different explanatory variables on lane deviation requires the use of sophisticated statistical and machine learning models, such as random forest (RF). RF feature selection is widely used in transportation research for its strength in handling complex, nonlinear relationships and multicollinearity among variables (12, 13). For example, Islam et al. (14) ranked the main contributors to pedestrian crash injury severity using RF importance measures. Guo et al. (15) confirmed RF's feature importance value in prioritizing both roadway and environmental features in crash models. Zhang et al. (16) combined RF feature selection with deep learning for real-time crash risk assessment, achieving better results than traditional methods. These studies highlight that RF importance measures, mean decrease in accuracy (MDA), and Gini-based metrics reliably identify the most important predictors (15, 17). While the RF model is effective for identifying important predictors and understanding their relative influence, it does not provide interpretable parameter estimates or account for repeated observations from the same driver. Because lane deviation values are nominal and the data structure contains multiple observations per driver, a multinomial mixed-effects model is needed to quantify effects while properly addressing within-driver correlation.

Multinomial mixed-effect models are popular because they improve on traditional multinomial logistic regression by including random effects to handle unobserved variability and correlations in clustered data (18, 19). Gkritza and Mannering (20) first used the mixed logit models in transportation safety by analyzing seatbelt use, accounting for behavioral heterogeneity across demographic groups. Since then, numerous subsequent studies have continued utilizing mixed logit and other random-parameter modeling approaches (21, 22). Song et al. (23) applied hierarchical multinomial logistic regression for crash severity on rural highways. Bham et al. (24) used mixed-effects multinomial regression with detailed roadway and driver data to reveal risk factors for single-vehicle crashes, showing how random effects capture hidden differences across segments in Arkansas.

Savolainen et al. (25) compared multinomial logit, ordered probit, and mixed logit models, highlighting that mixed logit models offer superior flexibility in addressing unobserved heterogeneity. Islam et al. (14) utilized hierarchical multinomial logit models to assess signal strategy impacts on crash severity, incorporating random

effects for intersection-level variations. Lee et al. (26) introduced mixed fractional split multinomial logit models with spatial effects to study vehicle-type crash proportions, integrating spatially correlated random effects for neighborhood-level influences. Together, these studies highlight the importance of employing flexible modeling approaches that account for heterogeneity and thus enabling more accurate evaluation of explanatory factors.

This study aims to investigate the influence of separation type on the driver's lane position on ML facilities. It focuses on the vehicles traveling on the ML adjacent to the separation type. The findings of this study can provide valuable insights for transportation agencies concerning the performance of different separation types on ML facilities.

Data Description

This study utilized a naturalistic driving study (NDS) to examine the impact of separation type on lane deviation along ML facilities. NDS data captures real-world driving behavior and safety-critical events as drivers operate in their natural, everyday environment. This makes NDS data particularly valuable, as it allows researchers to study near-crash situations that are difficult to capture in simulations or through reported data, such as traditional crash databases.

The NDS data used in this study were obtained from the Strategic Highway Research Program 2 (SHRP2), which is managed by the Virginia Tech Transportation Institute. The SHRP2 program represents the largest and most comprehensive NDS ever undertaken, involving approximately 3,400 participant drivers over three years. Data were collected from six states—Washington, Florida, New York, North Carolina, Pennsylvania, and Indiana—providing a diverse set of driving environments and behaviors.

Within the SHRP2 dataset, in-vehicle information was recorded using a data acquisition system (DAS). The DAS captured a wide range of kinematic data, including speed, acceleration, and deceleration, as well as video footage from four camera angles: front, rear, driver's face, and over the driver's shoulder. Particularly significant for this study, the dataset also included measurements of lane offset from the centerline, which formed the basis for calculating lane deviation values. In addition, SHRP2 data included driver demographic information, such as age and gender.

For this analysis, SHRP2 data focusing on ML facilities along the I-5 and I-90 freeways in Seattle, Washington, were used to examine the impact of ML separation type on lane deviation. Because the study focused specifically on the effect of separation type, the

Table 1. Summary Statistics of the Trips in SHRP2

Separation type	Events	Observations	Lane deviation	
			Mean (ft)	SD (ft)
Buffer	3,362	120,542	−0.445	1.33
Concrete barrier	502	99,482	−0.237	1.34
Median	2,112	41,335	−0.532	1.46
Total	5,976	261,359	na	na

Note: SD = standard deviation; SHRP2 = Strategic Highway Research Program 2; na = not applicable.

dataset was filtered to include only events in which the subject vehicles were traveling in the inside lane of the ML, that is, the lane directly adjacent to the separation type. During the filtering process, the SHRP2 dataset's trip-level images were reviewed to verify that the vehicles were indeed traveling in the inside lane of the ML facility.

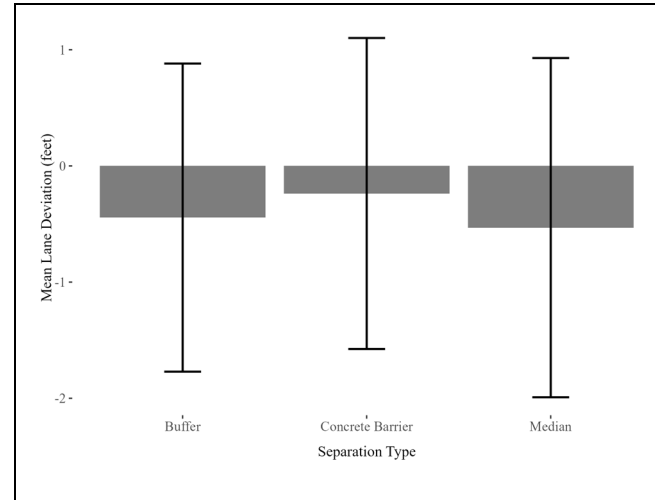
A total of 5,976 events met the criteria for inclusion, yielding 265,981 individual data points. Each data point corresponds to one second (1 s) of vehicle travel recorded during the event. These observations came from 310 participant drivers. Additional data on geometric characteristics were collected from SHRP2 images, the Geographic Information System database, and Google Maps.

Descriptive Statistics

Lane Deviation

The response variable for this study was lane deviation. Lane deviation values were obtained directly from the time-series data. The time-series data consists of second-by-second records of various driving parameters captured throughout each trip or event. Lane deviation values were recorded as either positive or negative based on a vehicle's position relative to the lane centerline. A positive value indicates that a vehicle's centroid was to the right of the lane centerline, while a negative value indicates it was to the left. Lane deviation values, defined as the offset between the lane center and the vehicle's centroid ranged from −7 ft to +7 ft. Of the 261,359 data points analyzed, 120,542 (46.1%) were observed on MLs with buffer separation, 38.1% on facilities with concrete barriers, and 15.8% on facilities with median separation. Table 1 summarizes the lane deviation values and the corresponding trips. Figure 3 presents the mean and standard deviation of lane deviation by separation types.

Figure 3 presents the graphical representation of the mean and standard deviation for different separation types. Based on Figure 3, on average, the drivers in the ML tend to move away from the separator for all

**Figure 3.** Mean lane deviation by separation type.

separation types, as observed by the negative values of the mean lane deviation. However, the magnitude varies across the separation types, with drivers on median-separated MLs tending to drive further away from the separator than with a concrete barrier or buffer separation.

Other Explanatory Variables

Table 2 presents the descriptive statistics of vehicle, roadway, and driver characteristics variables analyzed for the inside lane of the MLs. As shown in Table 2, lane deviation variable was categorized into three groups: between −0.5 and 0.5 ft (35.9%), greater than 0.5 ft (20.0%), and less than −0.5 ft (44.1%). In relation to separation type, over 38% of the data corresponded to facilities with the concrete barrier, while facilities with the median and buffer separation accounted for 15.8% and 46.1%, respectively.

The majority of the data points were generated by passenger cars (61.5%), compared with SUVs, which accounted for 38.5% of all data points. Single-lane ML facilities were observed in 54.9% of the data points, while facilities with two or more MLs accounted for 45.1%. When considering GPLs, over 39.9% of facilities had three lanes, while about 60.1% had four or more lanes.

Concerning driver-related characteristics, 65.9% of the recorded observations involved male drivers, while 34.1% involved female drivers. Most drivers were 20–39 years old (41.4%) or 40–65 years old (42.3%), with smaller proportions involving teenage drivers (5.7%) and those over 65 (10.6%). Drivers identifying as Hispanic/Latino represented 12.8% of the sample, while 87.2% were non-Hispanic/Latino.

Table 2. Descriptive Statistics

Variables	Levels	Count	Percentage
Response variable			
Lane deviation ^a	–0.5 ft to 0.5 ft	93,758	35.9
	Greater than 0.5 ft	52,177	20.0
	Less than –0.5 ft	115,424	44.1
Vehicle characteristics			
Vehicle class	Passenger car	160,679	61.5
	SUV	100,420	38.5
Roadway characteristics			
Separation type	Buffer	120,542	46.1
	Concrete barrier	99,482	38.1
	Median	41,335	15.8
Number of lanes—ML	1	143,480	54.9
	2	117,879	45.1
Number of lanes—GPL	3	104,439	39.9
	4+	156,920	60.1
Driver characteristics			
Driver gender	Male	172,365	65.9
	Female	88,994	34.1
Driver age group	< 20 years	14,866	5.7
	20–39 years	108,255	41.4
	40–65 years	110,507	42.3
	65+ years	27,731	10.6
Ethnicity	Hispanic/Latino	33,377	12.8
	Not Hispanic/Latino	227,982	87.2
Vision acuity—far	Exactly 20/40	10,105	3.9
	< 20/40	1,025	0.4
	> 20/40	250,229	95.7
Vision acuity—near	Exactly 20/40	9,315	3.6
	< 20/40	8,811	3.4
	> 20/40	243,233	93.0
Miles driven last year	Between 6,000 and 10,000	83,696	32.0
	Less than 6,000	24,880	9.5
	Greater than 10,000	152,783	58.5
Risk-taking behavior (secondary tasks)	Less risk	38,865	14.9
	Moderate risk	24,175	9.2
	High risk	198,319	75.9
Continuous variables			
Shoulder width (feet)	Mean	4.8	na
	Standard deviation	4.7	na
	Maximum	16	na
	Minimum	0	na
Lane width (feet)	Mean	11.0	na
	Standard deviation	4.6	na
	Maximum	15	na
	Minimum	7	na

Note: SUV = sports utility vehicle; ML = managed lanes; GPL = general-purpose lanes; na = not applicable.

^aRepresents the response variable.

Vision acuity data revealed that 95.7% of drivers had far vision acuity better than 20/40, while only 3.9% had visual acuity worse than 20/40. Similarly, for near vision acuity, 93.0% of drivers had a visual acuity better than 20/40, and 3.4% had a visual acuity worse than 20/40. Based on the total miles driven, over 58.5% of trips were made by drivers who drove more than 10,000 miles in the last year, while about 9.5% of trips were made by drivers who drove less than 6,000 miles last year. Over 75.9% of

the drivers categorized themselves as higher risk individuals when it comes to engaging in secondary tasks, such as phone usage or eating, that tend to distract drivers from the driving task.

Methodology

This section outlines the methodology adopted in this study to investigate the impact of separation type and

other explanatory variables on lane deviation on the inside lane of the MLs (i.e., the ML adjacent to the separator). The methodology was divided into three steps. The first step involves the use of an Analysis of Variance (ANOVA). This step was necessary because the study focuses on the influence of separation type. ANOVA enables assessment of whether the mean lane deviation differs significantly across the three separation types. The second step applies feature selection using a RF model to identify which explanatory variables contribute most to the variability in the response variable. Finally, a mixed-effects multinomial model was employed to quantify the impact of each variable on lane deviation in MLs. A detailed explanation of each step is provided in the following subsections.

Analysis of Variance

ANOVA was used to assess whether the type of separation significantly affected vehicle lateral position. By comparing these variances, ANOVA assesses whether the group means are statistically significantly different from each other. Equation 1 presents the fundamental ANOVA formula (27, 28):

$$\sum_{i=1}^m \sum_{j=1}^n (X_{ij} - \bar{X}_{..})^2 = \sum_{i=1}^m \sum_{j=1}^n (X_{ij} - \bar{X}_i)^2 + n \sum_{i=1}^m (X_i - \bar{X}_{..})^2 \quad (1)$$

where X_{ij} = observed lane deviation for the j^{th} observation in the i^{th} group; $\bar{X}_{..}$ = grand mean of all observations; and $\bar{X}_i$ = mean lane deviation of the i^{th} group. In addition, the Total Sum of Squares (SST), represented as $\sum_{i=1}^m \sum_{j=1}^n (X_{ij} - \bar{X}_{..})^2$, captures the overall variation in the data. The Error Sum of Squares (SSE), expressed as $\sum_{i=1}^m \sum_{j=1}^n (X_{ij} - \bar{X}_i)^2$, reflects the variation within each group, while the Treatment Sum of Squares (SS(Tr)), defined as $n \sum_{i=1}^m (X_i - \bar{X}_{..})^2$, measures the variation between groups.

The F -test statistic is computed by comparing the mean squared treatment variance to the mean squared error variance, as shown in Equation 2 (29, 30):

$$F = \frac{SS(Tr)/(m-1)}{SSE/(m(n-1))} = \frac{MS(Tr)}{MSE} \sim F[m-1, m(n-1)] \quad (2)$$

where $MS(Tr)$ = Mean Squared Treatment; MSE = Mean Squared Error; m = number of groups; and n = number of observations per group.

The null hypothesis (H_0) states that there are no differences in the population means of lane deviation values

across different separation types (Equation 3). The alternative hypothesis states that at least two of the population means of lane deviation values are different (Equation 4):

$$H_0 : \mu_1 = \mu_2 = \mu_3 \quad (3)$$

$$H_a : \text{At least one group mean is different} \quad (4)$$

where μ_1 = the mean lane deviation for separation type 1 (buffer separation); μ_2 = separation type 2 (median separation); and μ_3 = separation type 3 (concrete barrier separation). The alternative hypothesis asserts that at least two population means differ significantly.

Post-Hoc Analysis

Post-hoc analysis refers to statistical analyses that are conducted after an initial analysis has been completed, aiming to find patterns, trends, or differences within the data that were not specified in the earlier study. In this study, in case the null hypothesis is rejected in the ANOVA analysis, post-hoc analysis will be adopted to identify the specific differences between pairs of group means. This study utilized Tukey's Honest Significant Difference test for pairwise comparisons (31, 32). Tukey's test simultaneously evaluates all group mean pairs and identifies statistically significant differences exceeding the critical value based on the standard error (32). Equation 5 represents Tukey's test formula (33, 34):

$$D_a = \frac{U_P - V_Q}{SE} \quad (5)$$

where U_P = larger of the two means being compared; V_Q = smaller of the two means being compared; and SE = standard error associated with the difference between the means.

The critical Tukey test statistic D from the studentized range distribution is computed using Equation 6 (32, 35):

$$D = \frac{U_{max} - V_{min}}{S\sqrt{(2/n)}} \quad (6)$$

where U_{max} = largest sample mean among the groups; V_{min} = smallest sample mean among the groups; S = pooled standard deviation across groups; and n = number of observations per group.

The degree of freedom for each mean can be calculated as $N-k$, where N is the total number of observations and k is the number of subpopulations being compared.

Feature Selection Using RF

RF is a widely used machine learning algorithm that constructs a collection of decision trees and aggregates their

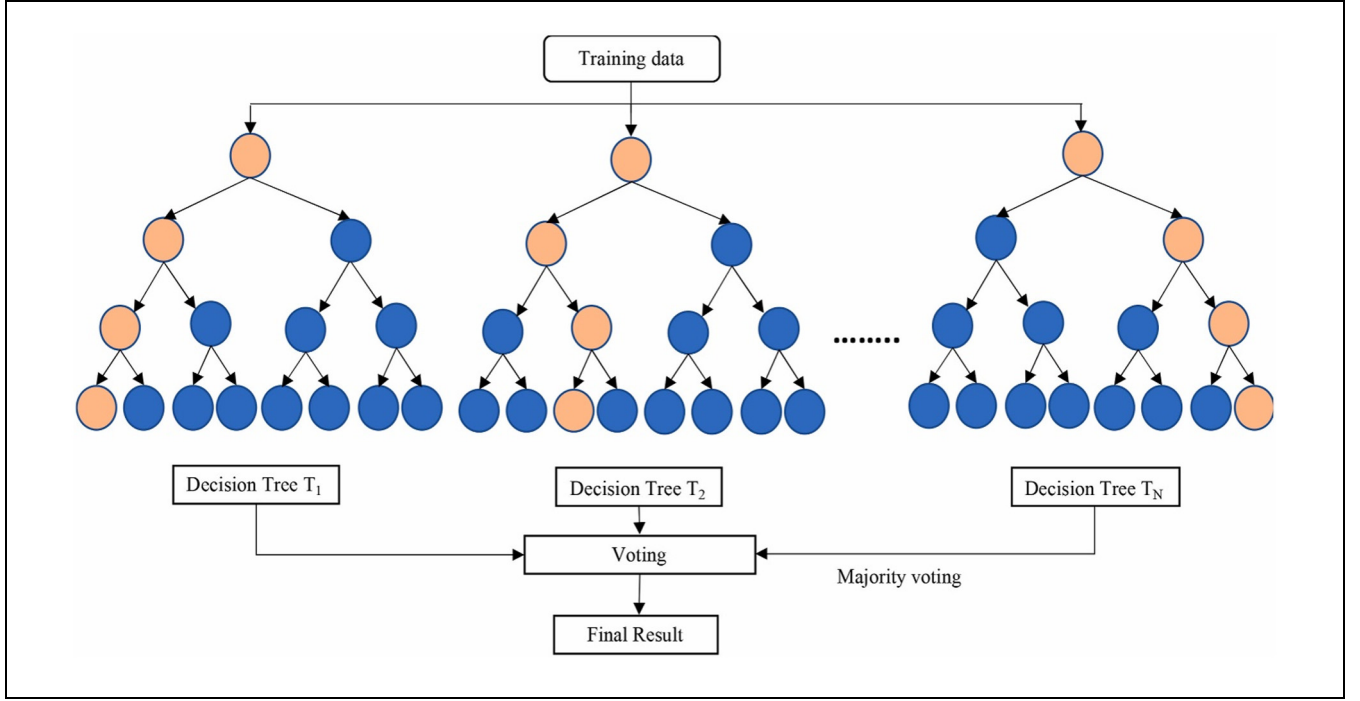


Figure 4. Random forest algorithm (37).

predictions for robust and accurate classification (36). As illustrated in Figure 4, each tree in the forest is trained on a bootstrap sample from the dataset, and at each node, a random subset of features is considered for splitting. This approach mitigates overfitting and improves generalizability, which is particularly important in transportation safety studies where predictor variables may be highly correlated or noisy (37, 38).

To ensure the reliability and robustness of feature importance estimates, key RF hyperparameters such as the number of trees ($n_{tree} = 500$), the number of features considered at each split ($m_{try} = \lfloor \sqrt{13} \rfloor$; based on the 13 predictors), and the maximum tree depth were systematically optimized using cross-validation (39). The tuning process followed a grid search strategy to balance model accuracy and computational efficiency, preventing bias in feature importance measures (40).

At the core of RF are decision trees. For a node t , the optimal split is determined by maximizing the decrease in an impurity measure such as the Gini impurity. Given class proportions $p_k(t)$ for class k at node t , the Gini impurity is defined in Equation 7 (38):

$$G(t) = 1 - \sum_{k=1}^K [p_k(t)]^2 \quad (7)$$

Each tree is grown to maximal depth or until a stopping criterion is met, and predictions (Equation 8) for each observation are aggregated across all trees using a majority vote (41):

$$\hat{y}_i = \text{mode}(\hat{y}_i^{(1)}, \hat{y}_i^{(2)}, \dots, \hat{y}_i^{(B)}) \quad (8)$$

where $\hat{y}_i^{(b)}$ is the prediction for observation i from tree b , and B is the total number of trees.

A key advantage of RF is its capacity to quantify feature importance (42). The MDA is calculated by permuting the values of each feature X_j in the out-of-bag (OOB) samples—those not used for fitting a particular tree—and measuring the change in model accuracy as shown in Equation 9 (43):

$$MDA_j = \frac{1}{B} \sum_{b=1}^B \left(\text{Accuracy}_{\text{OOB}}^{(b)} - \text{Accuracy}_{\text{OOB, perm}(j)}^{(b)} \right) \quad (9)$$

where $\text{Accuracy}_{\text{OOB}}^{(b)}$ is the OOB accuracy for tree b and $\text{Accuracy}_{\text{OOB, perm}(j)}^{(b)}$ is the accuracy after permuting X_j .

Mixed-Effect Multinomial Model

A mixed-effect multinomial logit model was used to analyze the three-category lane deviation outcome (-0.5 ft to 0.5 ft, greater than 0.5 ft, and less than -0.5 ft). This model incorporates both fixed effects (predictors measured for each observation) and random effects (to capture unobserved heterogeneity among participants), making it particularly appropriate for hierarchical or

Table 3. ANOVA Results

Factor	Df	Sum Sq	Mean Sq	F-value	P-value
Separation type	2	3480	1740.00	950.60	<0.001
Residuals	261356	478411	1.80	na	na

Note: Df = degrees of freedom; Sq = square; na = not applicable.

repeated-measures data commonly found in transportation research (44, 45).

Let Y_{ij} denote the lane deviation category for the i -th observation from the participant j ($Y_{ij} \in \{0, 1, 2\}$). The categories are defined as: 0, no significant deviation (-0.5 ft to 0.5 ft); 1, rightward deviation (greater than 0.5 ft); and 2, leftward deviation (less than -0.5 ft). The linear predictor for each nonreference category k ($k = 1, 2$) is defined as in Equation 10 (46):

$$\eta_{ijk} = \beta_{0k} + \mathbf{x}_{ij}^T \beta_k + u_{0k}^{(j)} \quad (10)$$

where β_{0k} = intercept for category k ; $\mathbf{x}_{ij}$ = vector of predictors for observation i of participant j ; β_k = vector of fixed-effect coefficients for category k ; and $u_{0k}^{(j)}$ = participant-specific random intercept for category k , assumed $u_{0k}^{(j)} \sim N(0, \sigma_{uk}^2)$.

With the three categories of the response variable, the multinomial logistic regression uses category 0 (no deviation) as the reference. The mixed-effect extension incorporates participant-level random effects to capture unobserved individual heterogeneity.

The probability of observing category k is expressed in Equation 11 (47):

$$P(Y_{ij} = k | \mathbf{x}_{ij}, u^{(j)}) = \frac{\exp(\eta_{ijk})}{1 + \sum_{l=1}^{K-1} \exp(\eta_{ijl})} \quad (11)$$

while the probability of the reference category 0 (no deviation) is given by Equation 12 (46):

$$P(Y_{ij} = 0 | \mathbf{x}_{ij}, u^{(j)}) = \frac{1}{1 + \sum_{l=1}^{K-1} \exp(\eta_{ijl})} \quad (12)$$

The inclusion of Participant ID as a random effect acknowledges that repeated lane deviation measurements from the same individual are correlated and allows for unobserved factors to influence the lane deviation (48). This structure yields more accurate effect estimates, and reliable standard errors compared with fixed-effects models, which ignore within-participant correlation and may underestimate uncertainty (49).

Table 4. Tukey Test of Significance

Separation type	Difference	P-value
Concrete barrier—buffer	0.075	<0.001
Median—buffer	0.081	<0.001
Median—concrete barrier	0.006	0.012

Results and Discussion

This section discusses the ANOVA test results, the RF analysis, and the mixed-effects multinomial model. Variables are reported only if they were found to be significant at the 95% confidence level.

ANOVA Results

ANOVA was conducted to check whether there is a significant difference in the mean lane deviation across different separation types. Table 3 presents the ANOVA results. Table 4 presents the results of the Tukey test of significance at a 95% confidence level. The conclusion from the ANOVA analysis was that there was a significant difference in the mean lane deviation between at least two separation types at a 95% confidence level. Since the null hypothesis was rejected, a post hoc analysis was required to analyze each pair separately to identify whether there was a significant difference between the pairs. Results from the Tukey test of significance revealed that there was a statistically significant difference in mean lane deviation values between ML facilities with median separation and concrete barrier separation, as well as between ML facilities with median separation and buffer separation types.

Feature Selection Results

Figure 5 presents the results of variable importance. Figure 5 presents each feature on the vertical axis and its MDA on the horizontal axis. The MDA metric quantifies each feature's contribution to the RF model's predictive performance by measuring the decrease in classification accuracy when that feature is randomly permuted (43). Features that exceed a specific threshold are regarded as important for modeling (50). In this study, all features demonstrated positive MDA scores (i.e., $MDA > 0$), highlighting their significance in predicting lane deviation behavior (43). This comprehensive feature importance validates the multifaceted nature of lane deviation prediction and supports the inclusion of the complete feature set in the predictive model (43).

Correlation Matrix

Before fitting the model, the correlation between variables was examined. This step is crucial because

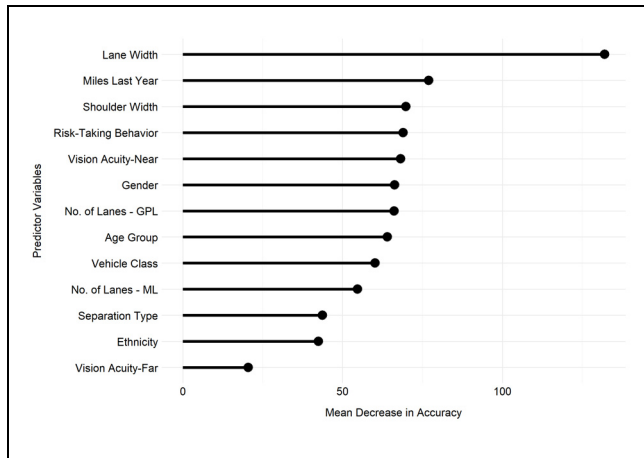


Figure 5. Variable selection.

Note: ML = managed lanes; GPL = general-purpose lanes.

including multiple highly correlated variables can be superfluous and can even impair the model's effectiveness. Figure 6 presents the correlation for the explanatory variables used in the model. The cutoff point selected for this study was 0.70 (51, 52). Based on the

results from Figure 6, no variables were highly correlated beyond the cutoff point.

Model Results

This subsection presents the results of the mixed-effect multinomial model. The purpose of using the multinomial model with mixed effects was to account for the unobserved heterogeneity in the data. The model was used to explore the influence of separation type and other explanatory variables on the lane deviation values. Table 5 provides the results of the multinomial logit model. As indicated in Table 5, 13 independent variables were significant at 95% confidence level. In the model, a negative coefficient suggests that the variable is associated with a reduction in the lane deviation offset. On the other hand, a positive coefficient indicates that the variable is associated with an increase in the lane deviation offset.

Effects of Separation Type on Lane Deviation. ML facilities with concrete barrier separation were associated with a 17% decrease in mean lane deviation away from the

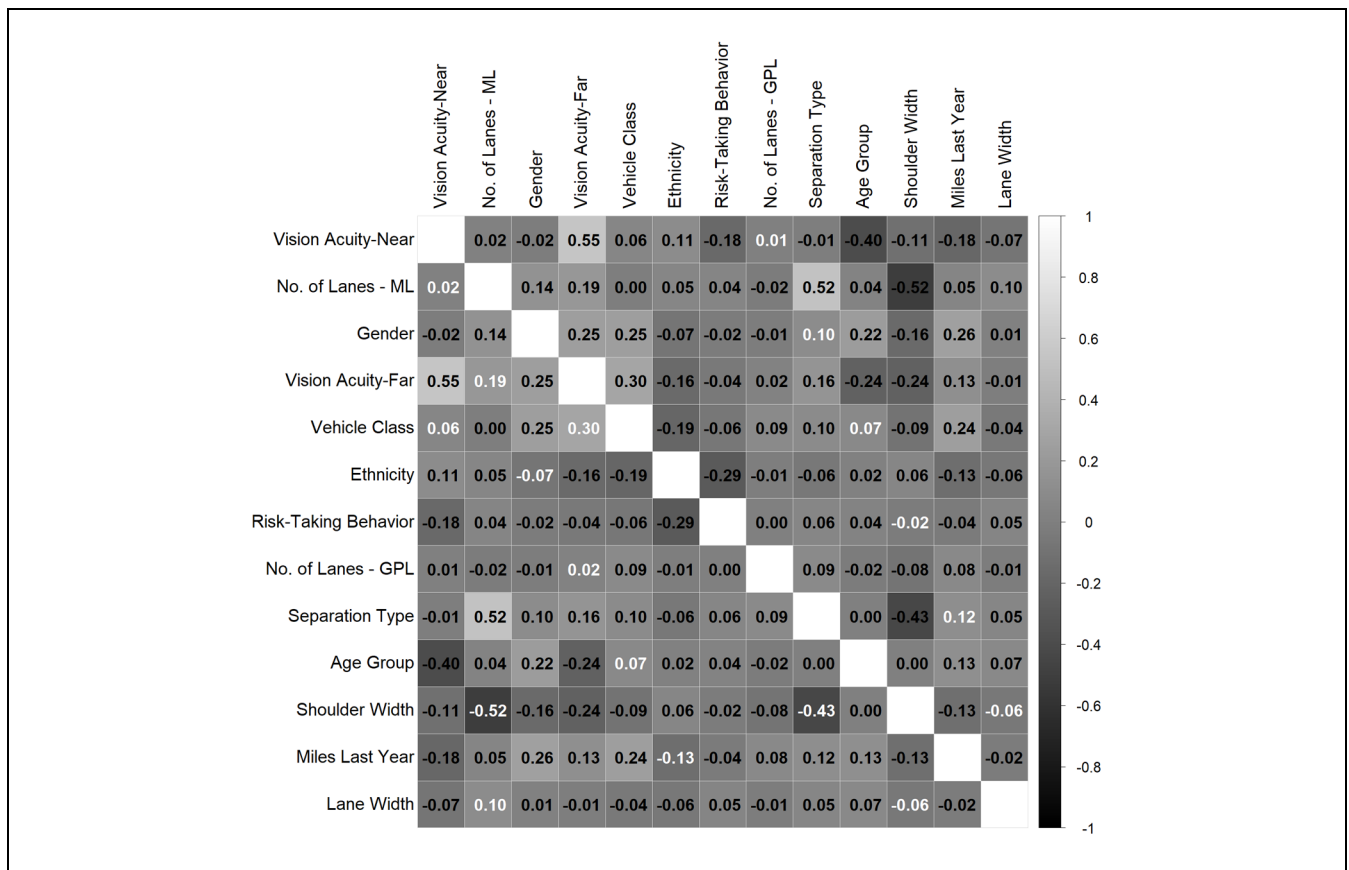


Figure 6. Correlation matrix.

Note: ML = managed lanes; GPL = general-purpose lanes.

Table 5. Mixed-Effect Multinomial Model Results

Variables	Factor 1	Estimate	SE	OR	P-value	Factor 2	Estimate	SE	OR	P-value
Intercept	Intercept 1	-3.173	0.066	0.04	<0.001	Intercept 2	-2.871	0.060	0.06	<0.001
Separation type	Buffer ^a					Buffer ^a				
	Concrete barrier	-0.186	0.037	0.83	<0.001	Concrete barrier	0.326	0.028	1.39	<0.001
	Median	-0.027	0.026	0.97	0.304	Median	0.388	0.019	1.47	<0.001
Gender	Female ^a					Female ^a				
	Male	-0.044	0.016	0.96	<0.001	Male	0.217	0.013	1.24	<0.001
Vehicle type	Passenger car ^a					Passenger car ^a				
	SUV	0.020	0.017	1.02	<0.001	SUV	-0.006	0.014	0.99	0.652
Ethnicity	Non-Hispanic or Latino ^a					Non-Hispanic or Latino ^a				
	Hispanic or Latino	0.100	0.021	1.11	<0.001	Hispanic or Latino	0.163	0.017	1.18	<0.001
Age group	20–40 ^a					20–40 ^a				
	<20	-0.576	0.039	0.56	<0.001	<20	-0.129	0.027	0.88	<0.001
	40–65	0.081	0.017	1.08	<0.001	40–65	0.098	0.014	1.10	<0.001
	65+	-0.061	0.025	0.94	0.924	65+	0.004	0.019	1.00	0.023
Lane width	Continuous	0.281	0.003	1.32	<0.001	Continuous	0.247	0.002	1.28	<0.001
Miles last year	(6,000, 10,000] ^a					(6,000, 10,000] ^a				
	< 6,000	-0.076	0.027	0.93	<0.001	< 6,000	-0.432	0.021	0.65	<0.001
	> 10,000	-0.187	0.028	0.83	<0.001	> 10,000	-0.462	0.021	0.63	<0.001
Risk-taking behavior—	Moderate risk ^a					Moderate risk ^a				
Secondary tasks	Less risk	0.112	0.029	1.12	<0.001	Less risk	0.135	0.016	1.14	<0.001
Far acuity in both eyes	High risk	0.103	0.021	1.11	<0.001	High risk	0.141	0.023	1.15	<0.001
	20/40 Only ^a					20/40 Only ^a				
	< 20/40	-0.007	0.039	0.99	0.850	< 20/40	-0.061	0.032	0.94	0.054
	> 20/40	0.033	0.133	1.03	0.803	> 20/40	1.182	0.099	3.26	<0.001
Near acuity in both eyes	20/40 Only ^a					20/40 Only ^a				
	< 20/40	-0.328	0.036	0.72	<0.001	< 20/40	0.920	0.038	2.51	<0.001
	> 20/40	0.079	0.054	1.08	0.144	> 20/40	1.141	0.051	3.13	<0.001
Number of lanes—	1 ^a					1 ^a				
ML	2+	0.127	0.035	1.14	<0.001	2+	-0.551	0.027	0.58	<0.001
Number of lanes—	3 ^a					3 ^a				
GPL	4+	-0.276	0.014	0.76	<0.001	4+	0.070	0.012	1.07	<0.001
Shoulder width	Continuous	-0.006	0.002	0.99	0.073	Continuous	-0.031	0.001	0.97	<0.001

Note: SE = standard error; OR = odds ratio; SUV = sports utility vehicle; ML = managed lanes; GPL = general-purpose lanes.

^aBase category; values in bold are significant at a 95% confidence level; Factor 1 corresponds to driving away from the separator; Factor 2 corresponds to driving toward the separator.

separator when compared with ML facilities with buffer separation. This suggests that, on average, drivers in MLs tend to drive farther away from the separator when the facility is designed with buffer separation than when it is constructed with concrete barriers. This observation could be the result of concrete barriers providing a physical division between the MLs and the adjacent GPLs. In addition, design standards for concrete barrier-separated facilities typically require the inclusion of a shoulder, which may give drivers a greater sense of safety and comfort, reducing the tendency to steer away from the separator.

When examining the impact of separation types on lane deviation toward the separator, ML facilities with both concrete barrier and median separation types were associated with an increase in mean lane deviation by 39% and 47%, respectively, toward the separator compared with facilities with buffer separation. This may be attributed to the physical separation present in both

concrete barrier and median designs, which can influence driver perception and lane positioning. These findings are consistent with observations reported in previous studies (53).

Effects of Other Contributing Factors on Lane Deviation

Driver Characteristics. Male drivers were associated with a 3% decrease in lane deviation away from the separator compared with female drivers. This observation may be ascribed to the aggressive driving behavior observed more in male drivers compared with female drivers. A contrary observation was made by a study by Das et al. (54), where female drivers were found to have better lane keeping ability.

According to Table 5, drivers who drove more than 10,000 mi in the last year were associated with an increase in the mean lane deviation away from the separator compared with those who drove between 6,000

and 10,000 mi. This observation may be attributed to higher mileage and experienced drivers having more exposure and increased confidence (55).

According to Table 5, drivers with near acuity worse than 20/40 were associated with a 28% decrease in the mean lane deviation toward the separator compared with those with near acuity of 20/40. This finding aligns with previous studies suggesting that driver's vision ability affects lane-keeping behavior (55).

Drivers categorized as engaging in high risk-taking behavior were associated with an 11% increase in mean lane deviation away from the separator compared with moderately risky drivers. This suggests that high-risk drivers, who are more likely to engage in secondary activities, may become distracted, leading to difficulty maintaining their vehicle's position at the center of the lane (56).

According to Table 5, teenage drivers were associated with a 44% decrease in mean lane deviation away from the separator compared with drivers aged 20–39 years. This could be attributed to teenage drivers feeling more comfortable driving closer to the separator. It may also relate to age-specific driving behaviors, such as heightened peripheral awareness and quicker reaction times among younger drivers. This observation is consistent with previous studies (8).

Geometric Characteristics. Lane width was found to be one of the most influential factors in the analysis. An increase in lane width was associated with a 32% increase in the mean lane deviation away from the separator. This may be attributed to wider lanes providing drivers with a perception of more space, prompting them to drift slightly farther from the separator. This observation is consistent with prior research, where wider lanes are linked to greater lateral variability (10).

The results show that an increase in the inside shoulder width was associated with a slight but statistically significant decrease in mean lane deviation away from the separator. This suggests that when more shoulder space is present, drivers feel more comfortable as the shoulder creates a buffer between the lane and the separator. This observation is consistent with previous findings that roadway design elements like shoulder width can influence driver behavior (6, 10).

According to Table 5, ML facilities with two or more lanes were associated with a 14% increase in mean lane deviation toward the separator compared with single-lane ML facilities. This may be attributed to drivers feeling more comfortable maneuvering within multilane managed facilities, leading to a slight shift in their lateral positioning. Conversely, facilities with four or more GPLs were associated with a 24% decrease in mean lane deviation away from the separator compared with

facilities with three GPLs. This could be because freeway sections with more adjacent GPLs create a sense of traffic density or visual complexity, prompting drivers in the ML to maintain a steadier, more central position and reduce deviation. This observation aligns with prior research showing that surrounding traffic conditions and roadway cross-section design affect driver lane-keeping behavior (11).

Conclusions

This study investigated the impacts ML separation type on lane deviation, a critical factor with direct safety implications in ML operations. Lane deviation measures how closely vehicles travel to lane boundaries and physical separators; excessive deviation or positioning too close to separators can elevate the risk of sideswipe crashes and collisions with barriers. Using naturalistic driving data from SHRP2 in Washington State, which included over 5,976 trips by 310 drivers, the study focused on three ML separation types along I-90 and I-5 in the Seattle area: median, buffer, and concrete barrier.

The findings revealed a consistent trend; across all three separation types, drivers tended to steer slightly away from the separator. Drivers on MLs with wide buffer separation exhibited the greatest lateral deviation compared with those driving on ML facilities with concrete barriers or buffer separation. The findings from the ANOVA revealed that there was a significant difference in the mean lane deviation across the three separation types. An RF model was applied to assess the importance of explanatory variables in predicting lane deviation, with lane width emerging as the most influential factor.

Further, results from the mixed-effect multinomial model showed that separation type, driver age, driver gender, total miles driven, vision acuity, lane width, shoulder width, driver risk-taking behavior, and the number of MLs and GPLs were significant factors influencing mean lane deviation at the 95% confidence level. Notably, MLs with concrete barriers were associated with a 17% decrease in lane deviation away from the separator compared with buffer-separated facilities, while facilities with median separation were associated with a 47% increase in lane deviation toward the separator relative to buffer separation.

These findings have clear safety implications: the farther drivers position themselves from the separator, the greater the risk of sideswipe crashes and other lateral conflicts. Therefore, cities and agencies designing or upgrading ML facilities must carefully consider separation type, as this study demonstrates significant differences in average lane deviation across different separation types. Selecting the appropriate separator not only influences driver comfort and behavior but also plays a pivotal

role in reducing crash risks, especially sideswipes, and improving overall safety on ML facilities.

This research provides transportation agencies with a better understanding of the effects of different ML separation treatments on driver behavior across all age groups. The findings will also enable state departments of transportation and local agencies to make informed decisions on appropriate separation treatments between the MLs and the GPLs, aimed at improving safety and mobility on ML facilities for drivers of all ages.

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Author Contribution Statement

The authors confirm contribution to the paper as follows: study conception and design: E. Mwambeleko, P. Alluri, and HM. Nayem; data collection: E. Mwambeleko, HM. Nayem; analysis and interpretation of results: E. Mwambeleko, HM. Nayem, and P. Alluri; draft manuscript preparation: E. Mwambeleko, P. Alluri, and HM. Nayem. All authors reviewed the results and approved the final version of the manuscript.

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